```
#include <stdio.h>
#include <stdlib.h>

struct ll_node {
   int data;
   struct ll_node *next;
   struct ll_node *prev;
};

struct ll_node* insert(struct ll_node *head, int val);
struct ll_node* del(struct ll_node* head, int val);
void display(struct ll_node *head);

void main() {

   struct ll_node *myList = NULL;
   int choice, value;

   do {
      printf("\n------ MENU ------\n");
      printf("1-Add Element\n");
      printf("2-Remove Element\n");
      printf("3-Show Linked List\n");
      printf("4-Quit\n");
      printf("Select option: ");
      scanf("%d", &choice);

      switch(choice) {
```

```
        case 1:
            printf("Input number to insert: ");
            scanf("%d", &value);
            myList = insert(myList, value);
            break;
        case 2:
            printf("Input number to remove: ");
            scanf("%d", &value);
            myList = del(myList, value);
            break;

        case 3:
            display(myList);
            break;

        case 4:
            printf("Program terminated.\n");
            break;

        default:
            printf("Wrong choice! Try again.\n");
    }

  } while(choice != 4);
}

struct ll_node* del(struct ll_node* head, int val) {
```

```
if (head == NULL) {
   printf("LIST IS EMPTY\n");
   return head;
}

struct ll_node *temp = head;

while (temp != NULL && temp->data != val)
   temp = temp->next;

if (temp == NULL) {
   printf("ELEMENT NOT FOUND\n");
   return head;
}
if (temp->prev == NULL) {
   head = temp->next;
   if (head != NULL)
      head->prev = NULL;
}
else {
   temp->prev->next = temp->next;

   if (temp->next != NULL)
      temp->next->prev = temp->prev;
}

free(temp);
```

```
    return head;
}

struct ll_node* insert(struct ll_node *head, int val) {

    struct ll_node *newNode = malloc(sizeof(struct ll_node));
    newNode->data = val;
    newNode->next = NULL;
    newNode->prev = NULL;
    if (head == NULL)
        return newNode;
    if (val < head->data) {
        newNode->next = head;
        head->prev = newNode;
        return newNode;
    }

    struct ll_node *temp = head;
    while (temp->next != NULL && temp->next->data < val)
        temp = temp->next;
    newNode->next = temp->next;
    newNode->prev = temp;

    if (temp->next != NULL)
        temp->next->prev = newNode;

    temp->next = newNode;
```

```
    return head;
}

void display(struct ll_node *head) {

    printf("\nLinked List Elements:\n");

    while (head != NULL) {
        printf("%d  ", head->data);
        head = head->next;
    }

    printf("\n");
}
```

```
------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 1
Input number to insert: 10

------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 1
Input number to insert: 30

------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 1
Input number to insert: 20

------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 3

Linked List Elements:
10  20  30
```

```
------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 2
Input number to remove: 10

------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 3

Linked List Elements:
20  30
```